

Section B

- 9 (a) to compare two potentials / voltages
output depends upon which is greater M1
A1 [2]
- (b) (i) resistance of thermistor = $2.5\text{ k}\Omega$
resistance of X = $2.5\text{ k}\Omega$ C1
A1 [2]
- (ii) at 5°C / at $< 10^\circ\text{C}$, $V^- > V^+$ M1
so V_{OUT} is -9 V A1
at 20°C / at $> 10^\circ\text{C}$, $V^- < V^+$ and V_{OUT} is $+9\text{ V}$ B1
 V_{OUT} switches between negative and positive at 10°C B1 [4]
(allow similar scheme if 20°C treated first)
- 10 (a) product of density (of medium) and speed of sound (in the medium) B1 [1]